

Homework 2

Due : March 13th at 11 : 59 pm

Deliverables. Submit a single PDF of your write-up to Gradescope *HW2 Write-Up*.

Start each problem on a new page.

Honor Code

Write and sign the following statement:

“I certify that all solutions in this document are entirely my own and that I have not looked at anyone else’s solution. I have given credit to all external sources I consulted.”

Signature: Jingliang Tu

1 Paper Questions

Overview

Large Language Models (LLMs) are capable of doing amazing things! But how do we evaluate their performance? Unlike standard classification tasks, evaluation of LLM outputs involves nuance, style, and human judgment. This homework will help you explore a recent approach to evaluation: **Chatbot Arena**.

You are encouraged to skim related works, but focus on the problems, current solutions, contributions, methods, and limitations. This assignment is not about memorizing details, but about developing the ability to *read research papers methodically*. Most research papers follow a common structure: they first motivate a **problem**, then describe **current solutions** and their limitations, followed by the **proposed solution and key insights**, the **methods** used, and finally a discussion of **limitations**. This homework is designed to help you practice and internalize this process of critical reading as you answer the questions. We have also provided pointers to relevant section that you might want to pay more attention to for each question.

- Paper: Chatbot Arena (arXiv:2403.04132)

For each question, we are looking for high-level concise answers, no need to write essays.

Chatbot Arena

- a. **[Problem]** What problem is Chatbot Arena aiming to solve? Why is evaluating generative models challenging compared to classification tasks? **[Section: 1. Introduction]**

Answer: Chatbot Arena aims to solve the problem of accurately and robustly evaluating the alignment and capabilities of Large Language Models (LLMs) based on true human preferences. Evaluating generative models is notoriously challenging because, unlike traditional classification tasks that have objective ground-truth labels, open-ended generation lacks a single "correct" answer. Instead, generation quality depends heavily on subjective human factors such as nuance, style, helpfulness, and safety, which are difficult to capture with automated static metrics.

- b. **[Current works]** What are current approaches to solving this problem and why are they not sufficient? **[Section: Related Works, LLM Benchmark]**

Answer: Current approaches heavily rely on static, ground-truth-based benchmarks (e.g., MMLU, HELM). These are insufficient because they are not open-ended and fail to capture the interactive nature of real-world human-LLM conversations. Furthermore, their static nature makes them highly susceptible to "data contamination," where test data inadvertently leaks into the models' training sets, leading to over-fitting and inaccurate capability assessments.

- c. **[Proposed solution]** What are the inputs and outputs of Chatbot Arena?

Answer:

- **Inputs:** A free-form user prompt and the corresponding text outputs generated side-by-side by two anonymous LLMs.
- **Outputs:**
 - *Per-battle:* A human preference vote selecting one of four options (Model A is better, Model B is better, Tie, or Both are bad).

- *Aggregated*: A global leaderboard ranking the models based on their estimated Bradley-Terry coefficients (translated into Elo-like scores), which represent their expected win rates.

d. **[Key insight / contributions]** Why does the Chatbot Arena evaluation approach address the issues of prior benchmarks? What are their main contributions?

Answer: By crowdsourcing blind, pairwise A/B testing, Chatbot Arena effectively eliminates brand bias and mimics real-world interactive usage. Its open-ended nature continuously updates the prompt distribution, heavily mitigating static data contamination. Their main contributions are: (1) Developing and launching the Chatbot Arena platform as an open-source evaluation framework. (2) Releasing a massive, diverse dataset of human preference votes and conversations. (3) Establishing a robust statistical evaluation pipeline utilizing the Bradley-Terry model. (4) Designing an efficient, active sampling algorithm that significantly improves sample efficiency during evaluation.

e. **[Method details]**

(i) What is a Bradley-Terry coefficient, and how is this used in the Chatbot Arena scoring?

Answer: A Bradley-Terry (BT) coefficient is a scalar parameter representing the latent "skill" or capability of a model. In Chatbot Arena, the probability of model i defeating model j is modeled as a logistic function of the difference in their coefficients: $P(i > j) = \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}}$. The system uses Maximum Likelihood Estimation (MLE) over the entire dataset of pairwise votes to globally fit these coefficients and rank the models.

(ii) **Section 4** of the Chatbot Arena paper mentions that the model scoring system of the original Chatbot Arena interface used Elo scores instead of Bradley-Terry coefficients because they are "better for the purpose of statistical estimation." Why are Bradley-Terry coefficients more appropriate for this setting?

Answer: Elo is an *online* rating system designed for dynamic settings (like chess) where it updates scores step-by-step after each match. It inherently assumes players' skills might evolve over time, making it highly sensitive to the order sequence of matches. Conversely, the BT model is an *offline* method that fits all match outcomes simultaneously to find the coefficients that best explain the entire dataset globally. Since LLM weights and capabilities are completely static during evaluation, the sequence-independent global estimation provided by the BT model is far more statistically efficient and appropriate.

f. **[Limitations]** What are some limitations of Chatbot Arena? What are general limitations of human preference benchmarks? **[Section: 8. Discussion]**

Answer: According to Section 8, the limitations of Chatbot Arena include: (1) The user base is potentially biased, as it primarily consists of LLM hobbyists and researchers, which may not represent the general population. (2) The data comes entirely from an online chat interface, which might not accurately reflect real-world usage in production environments or specialized domains. (3) The platform currently overlooks the safety aspects of the models, focusing primarily on helpfulness and preference. General limitations of human preference benchmarks include the high cost of scaling human evaluation and the inherent subjectivity that can misalign with objective factual correctness.

Other Practical Resources

Machine Learning Competitions: Kaggle

Kaggle hosts many ML competitions with a tier/medal system valued by industry. Despite being competitive, there's a strong culture of code sharing that accelerates learning.

Recommended Entry-Level Competition

- Titanic: Machine Learning from Disaster

Other Relevant Competitions

- Chatbot Arena (LMSYS)
- WSDM Cup Multilingual Chatbot Arena
- LLM Detect AI-Generated Text
- LLM Classification and Fine-Tuning

2 Sum of Residuals

Consider a dataset $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^{D+1}$ and $y_n \in \mathbb{R}$. Here we assume that X already contains a column of 1s. We model the relationship between x and y using linear regression with parameters $w \in \mathbb{R}^{D+1}$:

$$\hat{y} = x^\top w.$$

We have computed the solution w^* to the least squares regression problem:

$$w^* = \arg \min_w \frac{1}{2} \sum_{n=1}^N (y_n - x_n^\top w)^2.$$

The residual vector is then defined as:

$$r = y - Xw^*$$

- (a) (2 pts) Use the normal equation to show that: $X^\top r = 0$.

Answer: The least squares objective in matrix form is $\min_w \frac{1}{2} \|y - Xw\|_2^2$. Taking the gradient with respect to w and setting it to zero yields the normal equation:

$$\nabla_w \left(\frac{1}{2} (y - Xw)^\top (y - Xw) \right) = -X^\top (y - Xw) = 0$$

Since w^* is the minimizer, we have $X^\top (y - Xw^*) = 0$. Substituting the definition of the residual vector $r = y - Xw^*$, we directly get $X^\top r = 0$.

- (b) (2 pts) Use the presence of a bias term (encoded in the column of 1s in X) to show that the sum of the residuals is 0.

Answer: Because X contains a column of 1s (let's assume it is the first column), the first row of $X^\top$ is a row vector of all 1s: $[1, 1, \dots, 1]$. From part (a), we know $X^\top r = 0$. The first element of the resulting zero vector corresponds to the dot product of the first row of $X^\top$ and r :

$$[1, 1, \dots, 1] \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix} = \sum_{n=1}^N r_n = 0$$

Thus, the sum of the residuals is exactly 0.

- (c) (2 pts) Suppose we add a point (x_j, y_j) to our dataset with a very large $|y_j|$ but x_j is close to the mean of x . Predict qualitatively how w will change and explain why least squares is sensitive to such a point. How can you modify the loss to make the model more robust to outliers?

Answer:

- **How w changes:** The model parameters will shift significantly to accommodate the new point. Because x_j is near the mean, it won't drastically tilt the slope, but the bias term (the intercept) will be pulled heavily toward the direction of the large y_j .
- **Why it is sensitive:** Least squares minimizes the *squared* residuals. A very large $|y_j|$ produces an enormous squared penalty, so the optimizer disproportionately shifts the regression line towards the outlier to minimize this quadratic explosion.

- **How to fix it:** To make the model more robust, we can change the loss function so that it scales linearly (rather than quadratically) for large errors. Common robust loss functions include the Mean Absolute Error (L1 loss) or the Huber loss.

3 Weighted Linear Regression

Consider a dataset $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^{D+1}$ and $y_n \in \mathbb{R}$. Here we assume that X already contains a column of 1s. We model the relationship between x and y using linear regression with parameters $w \in \mathbb{R}^{D+1}$:

$$\hat{y} = x^\top w.$$

Suppose we have nonnegative weights $\{\alpha_n\}_{n=1}^N \geq 0$ for each observation and our goal is to minimize the weighted least squares objective:

$$L(w) = \sum_{i=1}^N \alpha_i (y_i - x_i^\top w)^2.$$

In this problem, we will derive the w^* that minimizes this loss.

- (a) (2 pts) Write the weighted least-squares loss function $L(w)$ in matrix form using the diagonal matrix:

$$A = \text{diag}(\alpha_1, \dots, \alpha_N) \in \mathbb{R}^{N \times N}.$$

Your answer should not contain any summations.

Answer:

$$L(w) = (y - Xw)^\top A (y - Xw)$$

- (b) (2 pts) Compute the gradient of the loss and express your answer in matrix form (no summations). You can do this by working with the summation form and taking partial derivatives or by using the following gradient identities:

$$\begin{aligned} \nabla_w (u^\top w) &= u \\ \nabla_w (w^\top A w) &= (A + A^\top)w \end{aligned}$$

Answer: First, expand the matrix expression for $L(w)$:

$$L(w) = y^\top A y - y^\top A X w - w^\top X^\top A y + w^\top X^\top A X w$$

Since A is a diagonal matrix, it is symmetric ($A = A^\top$). The term $y^\top A X w$ is a scalar, so it equals its transpose: $(y^\top A X w)^\top = w^\top X^\top A^\top y = w^\top X^\top A y$. Thus,

$$L(w) = y^\top A y - 2(X^\top A y)^\top w + w^\top (X^\top A X) w$$

Taking the gradient with respect to w :

$$\nabla_w L(w) = -2X^\top A y + (X^\top A X + (X^\top A X)^\top) w$$

Since $(X^\top A X)^\top = X^\top A^\top (X^\top)^\top = X^\top A X$, we get:

$$\nabla_w L(w) = -2X^\top A y + 2X^\top A X w$$

- (c) (3 pts) Set the gradient equal to zero, and solve for the closed-form solution for w^* that minimizes the weighted least squares objective. in terms of X , y , and A , assuming $X^\top AX$ is invertible.

Answer: Set the gradient to 0:

$$-2X^\top Ay + 2X^\top AXw^* = 0$$

$$X^\top AXw^* = X^\top Ay$$

Assuming $X^\top AX$ is invertible, we left-multiply by $(X^\top AX)^{-1}$:

$$w^* = (X^\top AX)^{-1}X^\top Ay$$

- (d) (3 pts) Under what condition is $X^\top AX$ guaranteed to be invertible? Can we derive a simpler condition when $\alpha_i > 0$?

Answer: For $X^\top AX$ to be invertible, it must be positive definite. Since A has non-negative diagonal elements ($\alpha_i \geq 0$), we can define $A^{1/2} = \text{diag}(\sqrt{\alpha_1}, \dots, \sqrt{\alpha_N})$. Then $X^\top AX = (A^{1/2}X)^\top (A^{1/2}X)$. This matrix is invertible if and only if the matrix $A^{1/2}X$ has full column rank.

When $\alpha_i > 0$ for all i , all diagonal entries of $A^{1/2}$ are strictly positive, meaning $A^{1/2}$ is itself an invertible matrix. Multiplying an invertible matrix by X preserves the rank of X . Therefore, the simpler condition when $\alpha_i > 0$ is simply that the design matrix X must have full column rank.

4 MLE vs MAP

In this problem, we explore the difference between the Maximum Likelihood Estimate (MLE) and the Bayesian Maximum A Posteriori (MAP) estimate. Suppose we have a coin with unknown bias Y . If we flip the coin we observe a Bernoulli random variable:

$$X \sim \text{Bernoulli}(Y) = p(X = x | Y) = Y^x(1 - Y)^{1-x}$$

The coin will land heads ($X = 1$) with probability Y and tails ($X = 0$) with probability $1 - Y$. We flip the coin (independently) N times to collect the dataset $\mathcal{D} = \{X_n\}_{n=1}^N$. We observe the coin lands heads $N_1 = \sum_{n=1}^N X_n$ times and tails $N_0 = N - N_1$ times.

For this problem, we will also introduce a prior on the probability Y that the coin lands heads. The Beta distribution is a common prior for probabilities. The Beta distribution has the density:

$$p(y | \alpha, \beta) = \frac{1}{B(\alpha, \beta)} y^{\alpha-1} (1-y)^{\beta-1}, \quad y \in (0, 1),$$

where $\alpha > 0$ and $\beta > 0$ are parameters of the distribution. The function $B(\alpha, \beta)$ is the normalization function

$$B(\alpha, \beta) = \int_0^1 y^{\alpha-1} (1-y)^{\beta-1} dy$$

of the Beta distribution called the Beta function. As a prior we will need to pick values for α and β . These should be based on our beliefs about the coin. Here we will use $\alpha = 3$ and $\beta = 3$:

- (a) (2 pts) Write the likelihood $p(\mathcal{D}|Y)$ of the data given Y .

Answer: Because the coin flips are independent, the likelihood is the product of individual probabilities:

$$p(\mathcal{D} | Y) = \prod_{n=1}^N Y^{X_n} (1 - Y)^{1-X_n} = Y^{N_1} (1 - Y)^{N_0}$$

- (b) (2 pts) Compute the derivative of the log-likelihood function with respect to Y and solve for the MLE estimate for Y .

Answer: The log-likelihood is $\ell(Y) = \ln(p(\mathcal{D} | Y)) = N_1 \ln(Y) + N_0 \ln(1 - Y)$. Taking the derivative with respect to Y :

$$\frac{d\ell}{dY} = \frac{N_1}{Y} - \frac{N_0}{1 - Y}$$

Set the derivative to zero:

$$\frac{N_1}{Y} = \frac{N_0}{1 - Y} \implies N_1(1 - Y) = N_0 Y \implies N_1 - N_1 Y = N_0 Y \implies Y(N_1 + N_0) = N_1$$

Thus, the MLE estimate is $Y_{\text{MLE}} = \frac{N_1}{N_1 + N_0}$.

- (c) (6 pts) Write the posterior distribution $p(Y|\mathcal{D})$. Express this posterior in the form of a Beta distribution. What are the parameters of this Beta distribution in terms of α , β , N_0 , and N_1 ?

Answer: By Bayes' Theorem, the posterior is proportional to the likelihood times the prior:

$$p(Y | \mathcal{D}) \propto p(\mathcal{D} | Y)p(Y) \propto (Y^{N_1} (1 - Y)^{N_0}) \cdot (Y^{\alpha-1} (1 - Y)^{\beta-1})$$

$$p(Y | \mathcal{D}) \propto Y^{N_1 + \alpha - 1} (1 - Y)^{N_0 + \beta - 1}$$

This clearly takes the form of a Beta distribution. Therefore, $p(Y | \mathcal{D}) \sim \text{Beta}(\alpha', \beta')$, where the new parameters are:

$$\alpha' = N_1 + \alpha \quad \text{and} \quad \beta' = N_0 + \beta$$

Using our specific prior values $\alpha = 3, \beta = 3$, the parameters are $\alpha' = N_1 + 3$ and $\beta' = N_0 + 3$.

- (d) (4 pts) Compute the derivative of the log-posterior and solve for the MAP Estimate.

Answer: The log-posterior (ignoring normalization constants) is:

$$\log p(Y | \mathcal{D}) = (N_1 + \alpha - 1) \ln(Y) + (N_0 + \beta - 1) \ln(1 - Y) + C$$

Substitute $\alpha = 3$ and $\beta = 3$:

$$\log p(Y | \mathcal{D}) = (N_1 + 2) \ln(Y) + (N_0 + 2) \ln(1 - Y) + C$$

Take the derivative with respect to Y :

$$\frac{d}{dY} \log p(Y | \mathcal{D}) = \frac{N_1 + 2}{Y} - \frac{N_0 + 2}{1 - Y}$$

Set to zero to find the MAP estimate:

$$\frac{N_1 + 2}{Y} = \frac{N_0 + 2}{1 - Y} \implies (N_1 + 2)(1 - Y) = (N_0 + 2)Y$$

$$N_1 + 2 - (N_1 + 2)Y = N_0Y + 2Y \implies Y(N_1 + N_0 + 4) = N_1 + 2$$

$$Y_{\text{MAP}} = \frac{N_1 + 2}{N_1 + N_0 + 4}$$

- (e) (2 pts) In a 1 sentence, what is the difference between the MLE and MAP estimates? Note: simply putting the your solutions to the previous parts is not sufficient.

Answer: While the MLE relies exclusively on the observed frequencies in the data to estimate the parameter, the MAP estimate incorporates our prior beliefs as "pseudo-counts" to regularize the estimate and prevent extreme predictions, especially when the sample size is small.

5 Estimating the Population of Grizzly Bears

An environmentalist, Amy, wants to estimate the number of grizzly bears roaming in a forest of British Columbia, Canada. She tracks $n = 20$ bears on her first visit to the forest, and marks each one with an electronic transmitter. A month later, she returns to the same forest and tracks $k = 15$ bears, and finds that only $x = 7$ of them have the transmitter on them. Assume that on each visit, she observes a uniformly random sample of bears. Note that the numbers of bears tracked during Amy's two visits, n and k , were chosen by her. The number x of bears she found with transmitter attached is her only observation.

- (a) (2 pts) Assuming Amy was equally likely to encounter any particular grizzly bear during her visits, what is the likelihood $\mathcal{L}(N; x)$ of the bear population N given her observation (i.e., the number of bears with transmitter observed) x ?

Answer: Since the sample is uniformly random without replacement, this follows a hypergeometric distribution. The likelihood is the probability of drawing exactly x marked bears out of k total drawn, from a population of N where n are marked:

$$\mathcal{L}(N; x) = \frac{\binom{n}{x} \binom{N-n}{k-x}}{\binom{N}{k}}$$

- (b) (5 pts) One way to estimate the bear population is to maximize the likelihood $\mathcal{L}(N; x)$. Derive the expression for the MLE estimate of the population $\hat{N}$ in terms of the number of bears tracked in both visits (parameters n, k) and the number of bears with transmitters found (observation x). *Hint:* calculus may not be easy, can we use likelihood ratios instead?

Answer: To find the maximum of $\mathcal{L}(N; x)$ with respect to the discrete integer N , we look at the ratio $\frac{\mathcal{L}(N; x)}{\mathcal{L}(N-1; x)}$ and find where it drops below 1.

$$\frac{\mathcal{L}(N; x)}{\mathcal{L}(N-1; x)} = \frac{\binom{n}{x} \binom{N-n}{k-x} / \binom{N}{k}}{\binom{n}{x} \binom{N-1-n}{k-x} / \binom{N-1}{k}}$$

Expand the binomial coefficients:

$$= \frac{\frac{(N-n)!}{(k-x)!(N-n-k+x)!}}{\frac{N!}{k!(N-k)!}} \times \frac{\frac{(N-1)!}{k!(N-1-k)!}}{\frac{(N-1-n)!}{(k-x)!(N-1-n-k+x)!}}$$

Simplify the factorials:

$$\begin{aligned} &= \frac{(N-n)!}{(N-1-n)!} \cdot \frac{(N-1)!}{N!} \cdot \frac{(N-k)!}{(N-1-k)!} \cdot \frac{(N-1-n-k+x)!}{(N-n-k+x)!} \\ &= \frac{(N-n)}{1} \cdot \frac{1}{N} \cdot \frac{(N-k)}{1} \cdot \frac{1}{(N-n-k+x)} = \frac{(N-n)(N-k)}{N(N-n-k+x)} \end{aligned}$$

We set this ratio ≥ 1 to see when likelihood increases:

$$\frac{N^2 - nN - kN + nk}{N^2 - nN - kN + Nx} \geq 1$$

Since N and the denominator are positive, cross-multiply:

$$N^2 - nN - kN + nk \geq N^2 - nN - kN + Nx \implies nk \geq Nx \implies N \leq \frac{nk}{x}$$

The likelihood increases as long as $N \leq nk/x$. Therefore, the maximum likelihood estimate is the largest integer satisfying this:

$$\hat{N} = \left\lfloor \frac{nk}{x} \right\rfloor$$

(c) (4 pts) What is Amy's MLE estimate $\hat{N}$ of the bear population?

Answer: Substituting the given values $n = 20$, $k = 15$, and $x = 7$:

$$\hat{N} = \left\lfloor \frac{20 \times 15}{7} \right\rfloor = \left\lfloor \frac{300}{7} \right\rfloor = \lfloor 42.857 \dots \rfloor = 42$$

Amy's MLE estimate of the bear population is 42 bears.

6 One Dimensional Mixture of Two Gaussians

Suppose we have a mixtures of two Gaussians in $\mathbb{R}$ that can be described by a pair of random variables (x, z) where x takes values in $\mathbb{R}$ and z takes value in the set $\{0, 1\}$. The joint-distribution of the pair (x, z) is given to us as follows:

$$\begin{aligned} z &\sim \text{Bernoulli}(1 - \beta), \\ (x|z = k) &\sim \mathcal{N}(\mu_k, \sigma_k^2), \quad k \in \{0, 1\}, \end{aligned}$$

We use $\vec{\theta}$ to denote the set of all parameters $\beta, \mu_0, \sigma_0, \mu_1, \sigma_1$.

- (a) (3 pts) Write down the expression for the joint likelihood $p(x, z = 0|\theta)$ and $p(x, z = 1|\theta)$. What is the marginal likelihood $p(x|\theta)$?

Answer: Since $z \sim \text{Bernoulli}(1 - \beta)$, we have $p(z = 1|\theta) = 1 - \beta$ and $p(z = 0|\theta) = \beta$. The joint likelihoods are:

$$\begin{aligned} p(x, z = 0 | \theta) &= p(x | z = 0, \theta)p(z = 0 | \theta) = \beta \mathcal{N}(x | \mu_0, \sigma_0^2) \\ p(x, z = 1 | \theta) &= p(x | z = 1, \theta)p(z = 1 | \theta) = (1 - \beta) \mathcal{N}(x | \mu_1, \sigma_1^2) \end{aligned}$$

The marginal likelihood $p(x | \theta)$ is obtained by summing out z :

$$p(x | \theta) = p(x, z = 0 | \theta) + p(x, z = 1 | \theta) = \beta \mathcal{N}(x | \mu_0, \sigma_0^2) + (1 - \beta) \mathcal{N}(x | \mu_1, \sigma_1^2)$$

- (b) (3 pts) What is the log-likelihood $\ell(\vec{x}|\theta)$? Why is this hard to optimize?

Answer: For a set of i.i.d. points $\vec{x} = \{x_1, \dots, x_N\}$, the log-likelihood is:

$$\ell(\vec{x} | \theta) = \sum_{i=1}^N \log (\beta \mathcal{N}(x_i | \mu_0, \sigma_0^2) + (1 - \beta) \mathcal{N}(x_i | \mu_1, \sigma_1^2))$$

This is hard to optimize because we have the logarithm of a sum inside the summation. This blocks the log function from directly acting on the exponential within the Gaussian density. Consequently, the parameters cannot be decoupled, making the objective highly non-convex with many local optima, meaning no closed-form analytical solution exists (which is why algorithms like Expectation-Maximization are typically used).

7 A Bayesian Interpretation of Lasso

Suppose you are aware that the labels $\{y_i\}_{i=1}^n$ corresponding to sample points $\{\vec{x}_i\}_{i=1}^n \in \mathbb{R}^d$ follow the density law

$$f(y_i | \vec{x}_i, \vec{w}) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(y_i - \vec{w}^\top \vec{x}_i)^2}{2\sigma^2}\right)$$

where $\sigma > 0$ is a known constant and $\vec{w} \in \mathbb{R}^d$ is a random parameter. Suppose further that experts have told you that

- each component of $\vec{w}$ is independent of the others, and
- each component of $\vec{w}$ has the Laplace distribution with location 0 and scale being a known constant b . That is, each component w_i obeys the density law $f(w_i) = e^{-|w_i|/b}/(2b)$.

Assume the outputs $y_{i \in [n]}$ are independent from each other.

Your goal is to find the choice of parameter $\vec{w}$ that is *most likely* given the input-output examples $(\vec{x}_i, y_i)_{i \in [n]}$. This method of estimating parameters is called *maximum a posteriori* (MAP); Latin for “*maximum [odds] from what follows*.”

- (a) (6 pts) Derive the *posterior* probability density law $f(\vec{w} | \{(\vec{x}_i, y_i)\}_{i=1}^n)$ for $\vec{w}$ up to a *proportionality constant* by applying Bayes’ Theorem and substituting for the densities $f(y_i | \vec{x}_i, \vec{w})$ and $f(\vec{w})$. Don’t try to derive an exact expression for $f(\vec{w} | \{(\vec{x}_i, y_i)\}_{i=1}^n)$, as the denominator is very involved and irrelevant to maximum likelihood estimation.

Answer: Using Bayes’ Theorem, the posterior is proportional to the likelihood times the prior. Given that samples are independent and weight components are independent:

$$f(\vec{w} | \{(\vec{x}_i, y_i)\}_{i=1}^n) \propto f(\{(y_i, \vec{x}_i)\}_{i=1}^n | \vec{w}) f(\vec{w}) = \left(\prod_{i=1}^n f(y_i | \vec{x}_i, \vec{w}) \right) \left(\prod_{j=1}^d f(w_j) \right)$$

Substituting the given densities:

$$\propto \left(\prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(y_i - \vec{w}^\top \vec{x}_i)^2}{2\sigma^2}\right) \right) \left(\prod_{j=1}^d \frac{1}{2b} \exp\left(\frac{-|w_j|}{b}\right) \right)$$

We can drop the constants $\frac{1}{\sigma\sqrt{2\pi}}$ and $\frac{1}{2b}$ since we only need the expression up to a proportionality constant:

$$f(\vec{w} | \{(\vec{x}_i, y_i)\}_{i=1}^n) \propto \exp\left(-\sum_{i=1}^n \frac{(y_i - \vec{w}^\top \vec{x}_i)^2}{2\sigma^2} - \sum_{j=1}^d \frac{|w_j|}{b}\right)$$

- (b) (2 pts) Define the log-likelihood for MAP as $\ell(\vec{w}) := \ln f(\vec{w} | \{(\vec{x}_i, y_i)\}_{i=1}^n)$. Show that maximizing the MAP log-likelihood over all choices of $\vec{w}$ is the same as minimizing $\sum_{i=1}^n (y_i - \vec{w} \cdot \vec{x}_i)^2 + \lambda \|\vec{w}\|_1$ where $\|\vec{w}\|_1 = \sum_{j=1}^d |w_j|$ and λ is a constant. Also give a formula for λ as a function of the distribution parameters.

Answer: Taking the natural log of the unnormalized posterior derived in (a):

$$\ell(\vec{w}) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \vec{w}^\top \vec{x}_i)^2 - \frac{1}{b} \sum_{j=1}^d |w_j| + C$$

where C incorporates all the dropped constants. Maximizing $\ell(\vec{w})$ is equivalent to minimizing its negative:

$$\arg \max_{\vec{w}} \ell(\vec{w}) = \arg \min_{\vec{w}} \left(\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \vec{w}^\top \vec{x}_i)^2 + \frac{1}{b} \|\vec{w}\|_1 \right)$$

Since scaling the objective by a positive constant does not change the optimal $\vec{w}$, we can multiply the entire objective by $2\sigma^2$:

$$= \arg \min_{\vec{w}} \left(\sum_{i=1}^n (y_i - \vec{w}^\top \vec{x}_i)^2 + \frac{2\sigma^2}{b} \|\vec{w}\|_1 \right)$$

This exactly matches the form $\sum_{i=1}^n (y_i - \vec{w}^\top \vec{x}_i)^2 + \lambda \|\vec{w}\|_1$. The formula for the regularization constant λ is $\lambda = \frac{2\sigma^2}{b}$.

8 ℓ_1 -regularization, ℓ_2 -regularization, and Sparsity

You are given a design matrix X (whose i^{th} row is sample point $\vec{x}_i^\top$) and an n -vector of labels $\vec{y} \triangleq [y_1 \ \dots \ y_n]^\top$. For simplicity, assume $X^\top X = nI$. Do not add a fictitious dimension/bias term; for input $\vec{0}$, the output is always 0. Let $\vec{x}_{*i}$ denote the i^{th} column of X .

- (a) (4 pts) Show that the cost function for ℓ_1 -regularized least squares,

$$J_1(\vec{w}) := \|X\vec{w} - \vec{y}\|^2 + \lambda \|\vec{w}\|_1$$

(where $\lambda > 0$), can be rewritten as

$$J_1(\vec{w}) = \|\vec{y}\|^2 + \sum_{i=1}^d f(\vec{x}_{*i}, \vec{w}_i)$$

where $f(\cdot, \cdot)$ is a suitable function whose first argument is a vector and second argument is a scalar, and $\vec{w} \in \mathbb{R}^d$ is a d -dimensional weight vector.

Answer: Let's expand the squared error term:

$$\begin{aligned} J_1(\vec{w}) &= (X\vec{w} - \vec{y})^\top (X\vec{w} - \vec{y}) + \lambda \|\vec{w}\|_1 \\ &= \vec{w}^\top X^\top X \vec{w} - 2\vec{y}^\top X\vec{w} + \|\vec{y}\|^2 + \lambda \|\vec{w}\|_1 \end{aligned}$$

We are given $X^\top X = nI$. Substituting this in:

$$\begin{aligned} &\vec{w}^\top (nI) \vec{w} - 2(X^\top \vec{y})^\top \vec{w} + \|\vec{y}\|^2 + \lambda \|\vec{w}\|_1 \\ &= n\|\vec{w}\|^2 - 2 \sum_{i=1}^d (\vec{x}_{*i}^\top \vec{y}) \vec{w}_i + \|\vec{y}\|^2 + \lambda \sum_{i=1}^d |\vec{w}_i| \\ &= \|\vec{y}\|^2 + \sum_{i=1}^d \left(n\vec{w}_i^2 - 2(\vec{x}_{*i}^\top \vec{y}) \vec{w}_i + \lambda |\vec{w}_i| \right) \end{aligned}$$

Therefore, the function is $f(\vec{x}_{*i}, \vec{w}_i) = n\vec{w}_i^2 - 2(\vec{x}_{*i}^\top \vec{y}) \vec{w}_i + \lambda |\vec{w}_i|$.

- (b) (8 pts) Using your solution to the previous part, derive necessary and sufficient conditions for the i^{th} component of the optimizer $\vec{w}^*$ of $J_1(\cdot)$ to satisfy each of these three properties: $w_i^* > 0$, $w_i^* = 0$, and $w_i^* < 0$.

Answer: Since $J_1(\vec{w})$ is separable for each $\vec{w}_i$, we can optimize $f_i(w_i) = nw_i^2 - 2(\vec{x}_{*i}^\top \vec{y}) w_i + \lambda |w_i|$ independently. We find the optimum by looking at the subgradient ∂f_i :

$$\partial f_i(w_i) = 2nw_i - 2\vec{x}_{*i}^\top \vec{y} + \lambda \text{sign}(w_i)$$

Setting $0 \in \partial f_i(w_i^*)$:

- **For $w_i^* > 0$:** $\text{sign}(w_i^*) = 1$, so $2nw_i^* - 2\vec{x}_{*i}^\top \vec{y} + \lambda = 0 \implies w_i^* = \frac{2\vec{x}_{*i}^\top \vec{y} - \lambda}{2n}$.
For this to be strictly positive, we must have $2\vec{x}_{*i}^\top \vec{y} - \lambda > 0$, which means $\vec{x}_{*i}^\top \vec{y} > \frac{\lambda}{2}$.
- **For $w_i^* < 0$:** $\text{sign}(w_i^*) = -1$, so $2nw_i^* - 2\vec{x}_{*i}^\top \vec{y} - \lambda = 0 \implies w_i^* = \frac{2\vec{x}_{*i}^\top \vec{y} + \lambda}{2n}$.
For this to be strictly negative, we must have $2\vec{x}_{*i}^\top \vec{y} + \lambda < 0$, which means $\vec{x}_{*i}^\top \vec{y} < -\frac{\lambda}{2}$.

- **For $w_i^* = 0$:** The subdifferential at zero is $[-1, 1]$, so $0 \in [-2\vec{x}_{*i}^\top \vec{y} - \lambda, -2\vec{x}_{*i}^\top \vec{y} + \lambda]$. This implies $-\lambda \leq -2\vec{x}_{*i}^\top \vec{y} \leq \lambda$, which simplifies to $-\frac{\lambda}{2} \leq \vec{x}_{*i}^\top \vec{y} \leq \frac{\lambda}{2}$, or $|\vec{x}_{*i}^\top \vec{y}| \leq \frac{\lambda}{2}$.
- (c) (3 pts) For the optimizer $\vec{w}^\#$ of the ℓ_2 -regularized least squares cost function $J_2(\vec{w}) \triangleq \|X\vec{w} - \vec{y}\|^2 + \lambda\|\vec{w}\|^2$ (where $\lambda > 0$), derive a necessary and sufficient condition for $\vec{w}_i^\# = 0$, where $\vec{w}_i^\#$ is the i th component of $\vec{w}^\#$.

Answer: Using the exact same expansion from part (a) (but replacing L1 with L2), we get $f_i(w_i) = nw_i^2 - 2(\vec{x}_{*i}^\top \vec{y})w_i + \lambda w_i^2$. This function is smooth, so we take the standard derivative and set it to zero:

$$\frac{df_i}{dw_i} = 2nw_i - 2\vec{x}_{*i}^\top \vec{y} + 2\lambda w_i = 0 \implies 2(n + \lambda)w_i = 2\vec{x}_{*i}^\top \vec{y} \implies w_i^\# = \frac{\vec{x}_{*i}^\top \vec{y}}{n + \lambda}$$

For $w_i^\# = 0$, we simply require the numerator to be zero. Since $n + \lambda > 0$, the necessary and sufficient condition is $\vec{x}_{*i}^\top \vec{y} = 0$.

- (d) (3 pts) A vector is called *sparse* if most of its components are 0. From your solution to part (b) and (c), which of $\vec{w}^*$ and $\vec{w}^\#$ is more likely to be sparse? Why?

Answer: The optimizer $\vec{w}^*$ (from ℓ_1 -regularization) is much more likely to be sparse. In ℓ_1 -regularization, a parameter w_i^* is forced precisely to zero over an entire interval of values: whenever $|\vec{x}_{*i}^\top \vec{y}| \leq \frac{\lambda}{2}$. If data is noisy, it's very probable that inner products fall inside this non-zero range. Conversely, in ℓ_2 -regularization, $w_i^\# = 0$ requires the exact strict equality $\vec{x}_{*i}^\top \vec{y} = 0$. In continuous space with real-world noise, landing exactly on 0 is a measure-zero event, meaning it almost never happens, so practically none of the ℓ_2 weights become exactly zero.

9 Mixed Feelings

In class, we learned about the k -means problem for clustering a distribution of data. Formally, our objective was

$$\min_{S=C_1 \cup \dots \cup C_k, \{c_1, \dots, c_k\}} \sum_k \sum_{x \in C_k} \|x - c_k\|_2^2$$

where we are simultaneously optimizing the cluster partition and our cluster centroids. We can further generalize this problem by incorporating the cluster partitions into a notion we call the “risk” R of the centroids, which measures how “bad” our centroids and assignments are. Our new objective becomes

$$\min_{c_1, \dots, c_k} R(c_1, \dots, c_k) = \min_{c_1, \dots, c_k} \sum_{i=1}^n \min_{1 \leq \ell \leq k} \phi(\|x_i - c_\ell\|_2)$$

where $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ is a continuous penalty function with minimum value $\phi(0) = 0$. This objective is a standard formulation of the generalized k -means problem (GKM). We can quickly recover the k -means objective seen in class by using $\phi(t) = t^2$.

This problem is inspired by Robust k -means: a Theoretical Revisit from NeurIPS 2016.

Breakdown of Quadratic ϕ

Assume $k = 1$ with our data $x \in \mathbb{R}$ generated from a contaminated distribution with density

$$p(x) = (1 - \epsilon)\mathcal{N}(x|0, \sigma^2) + \epsilon\mathcal{C}(x|x_0, \gamma)$$

where $\epsilon \in (0, 1)$ is the constant contamination fraction, $\mathcal{N}(x|0, \sigma^2)$ represents the Gaussian inliers, and $\mathcal{C}(x|x_0, \gamma)$ represents Cauchy heavy-tailed outliers (as seen in lecture). We’ll investigate conditions for breakdown of our standard k -means problem using this setup.

As a reminder, here is the density for a univariate Gaussian $\mathcal{N}(x|\mu, \sigma^2)$:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

and the density for a univariate Cauchy $\mathcal{C}(x|x_0, \gamma)$:

$$p(x) = \frac{1}{\pi} \left[\frac{\gamma}{(x - x_0)^2 + \gamma^2} \right]$$

- (a) (8 pts) We define risk as the expected loss over our entire data distribution. Using this and $\phi(t) = t^2$, write the expanded integral form of the risk $R(c)$ over our distribution of x for our single centroid c . Then, prove that $R(c)$ diverges for all $c \in \mathbb{R}$ under this specific data distribution.

Answer: The expected risk is defined as:

$$R(c) = \int_{-\infty}^{\infty} (x - c)^2 p(x) dx$$

Substitute the given mixture distribution:

$$R(c) = \int_{-\infty}^{\infty} (x - c)^2 [(1 - \epsilon)\mathcal{N}(x|0, \sigma^2) + \epsilon\mathcal{C}(x|x_0, \gamma)] dx$$

$$R(c) = (1 - \epsilon) \int_{-\infty}^{\infty} (x - c)^2 \mathcal{N}(x|0, \sigma^2) dx + \epsilon \int_{-\infty}^{\infty} (x - c)^2 \mathcal{C}(x|x_0, \gamma) dx$$

The first integral corresponds to the expected squared distance of a Gaussian random variable. Since $\mathbb{E}[X^2]$ for $\mathcal{N}(0, \sigma^2)$ is σ^2 and $\mathbb{E}[X] = 0$, this evaluates to $\sigma^2 + c^2$, which is finite. However, the second integral is over the Cauchy distribution. The Cauchy distribution has exceedingly heavy tails, meaning that even its first moment (mean) is undefined, and its second moment strictly diverges to infinity. Consequently, $\int (x - c)^2 \mathcal{C}(x|x_0, \gamma) dx = \infty$ for any real c . Because $\epsilon > 0$, the risk sum includes an infinite term, meaning $R(c) = \infty$ for all $c \in \mathbb{R}$.

- (b) (4 pts) We replace the Cauchy contamination with a finite-variance Dirac delta point mass $\delta_M(x)$ at an arbitrary distance $M > 0$. Recall that this distribution is given by

$$\delta_M(x) = \begin{cases} \infty, & x = M \\ 0, & x \neq M \end{cases}$$

Now, the density of our data is

$$p(x) = (1 - \epsilon) \mathcal{N}(x|0, \sigma^2) + \epsilon \delta_M(x)$$

Find the location of the optimal centroid c^* . *Hint:* c^* minimizes risk.

Note: Since the uncontaminated Gaussian's optimal centroid is 0, your solution will demonstrate that the quadratic penalty is globally sensitive to outliers. As $M \rightarrow \infty$, the centroid shifts unboundedly away from the original centroid. As an optional exercise, you can demonstrate that in the finite data case, the contamination point is $\frac{1}{n}$, meaning a single arbitrary outlier is enough to cause the centroid estimate to diverge to infinity.

Answer: Write the risk functional with the new Dirac mass distribution:

$$R(c) = (1 - \epsilon) \int_{-\infty}^{\infty} (x - c)^2 \mathcal{N}(x|0, \sigma^2) dx + \epsilon \int_{-\infty}^{\infty} (x - c)^2 \delta_M(x) dx$$

As derived in the previous part, the Gaussian integral equals $\sigma^2 + c^2$. By the sifting property of the Dirac delta function, the second integral evaluates to $(M - c)^2$.

$$R(c) = (1 - \epsilon)(\sigma^2 + c^2) + \epsilon(M - c)^2$$

To find the optimal centroid c^* , take the derivative with respect to c and set to zero:

$$\frac{dR(c)}{dc} = 2(1 - \epsilon)c - 2\epsilon(M - c) = 0$$

$$c - \epsilon c - \epsilon M + \epsilon c = 0 \implies c - \epsilon M = 0$$

Therefore, the optimal centroid is $c^* = \epsilon M$.

Robust k -Means

To address the sensitivity issues discovered in the quadratic penalty, we wish to introduce the robust k -means (RKM) formulation, allowing each observation x_i to take on an error term o_i , penalized by a function f_λ to encourage these errors to be zero. We will explore how we can construct a hard-thresholding behavior where the algorithm can ignore outliers outside a fixed radius surrounding the centroid. Assume our data is generated from a continuous density $p(x) = (1-\epsilon)p_{\text{in}}(x) + \epsilon p_{\text{out}}(x)$ for $x \in \mathbb{R}^d$. We seek the optimal centroid $c^* \in \mathbb{R}^d$ and inlier weighting function $w(x) \in [0, 1]$ that minimizes the expected squared error of the inliers. Additionally, this problem is constrained by the requirement that exactly $1 - \alpha$ of the probability mass is assigned to the inliers.

- (c) (8 pts) Construct a Lagrangian functional $\mathcal{L}(w, c, \lambda)$ for the weighted expected error optimization problem

$$\min_w \mathbb{E}_{x \sim p(x)} [w(x) \|x - c\|_2^2]$$

subject to the probability mass constraint

$$\int_{\mathbb{R}^d} w(x) p(x) dx = 1 - \alpha$$

Then, group the terms into a single integral and optimize the Lagrangian pointwise with respect to $w(x)$ for a fixed centroid c and fixed Lagrange multiplier $\lambda > 0$. Conclude that the optimal weighting will exclude data outside some radius from the centroid. *Note:* $0 \leq w(x) \leq 1$.

Hint: Take a look at the Wikipedia article for Lagrange multipliers if you need help getting started. *Hint:* Write the constraint $g(w) = 0$ using

$$g(w) = 1 - \alpha - \int_{\mathbb{R}^d} w(x) p(x) dx$$

Hint: Once you write the integral, can you find the optimal $w^*(x)$ for different cases of the integrand?

Answer: First, construct the Lagrangian functional $\mathcal{L}(w, c, \lambda)$ using the objective and the constraint $g(w)$:

$$\mathcal{L}(w, c, \lambda) = \int_{\mathbb{R}^d} w(x) \|x - c\|_2^2 p(x) dx + \lambda \left(1 - \alpha - \int_{\mathbb{R}^d} w(x) p(x) dx \right)$$

Group the terms containing $w(x)$ into a single integral:

$$\mathcal{L}(w, c, \lambda) = \int_{\mathbb{R}^d} w(x) p(x) (\|x - c\|_2^2 - \lambda) dx + \lambda(1 - \alpha)$$

We want to minimize this Lagrangian with respect to the weights $w(x) \in [0, 1]$ pointwise for every x . Since $p(x) \geq 0$, the sign of the integrand is entirely determined by the term $(\|x - c\|_2^2 - \lambda)$. To minimize the integral, we should set $w(x)$ as large as possible when this coefficient is negative, and as small as possible when it is positive:

- If $\|x - c\|_2^2 - \lambda < 0$, we set $w^*(x) = 1$.
- If $\|x - c\|_2^2 - \lambda > 0$, we set $w^*(x) = 0$.

This implies $w^*(x) = \mathbb{I}_{\|x-c\|_2^2 \leq \lambda}$. Therefore, the optimal weighting hard-thresholds and entirely excludes any data points that are further than a squared distance of λ (or radius $\sqrt{\lambda}$) from the centroid.

- (d) (5 pts) Differentiate $\mathcal{L}$ with respect to c to find the optimal centroid c^* given the optimal weights $w^*(x)$ and write it as an expectation.

Answer: Take the gradient of the Lagrangian with respect to c :

$$\begin{aligned}\nabla_c \mathcal{L} &= \nabla_c \int_{\mathbb{R}^d} w^*(x)p(x)\|x-c\|_2^2 dx = \int_{\mathbb{R}^d} w^*(x)p(x)\nabla_c \|x-c\|_2^2 dx \\ &= \int_{\mathbb{R}^d} w^*(x)p(x)(-2(x-c))dx = -2 \int_{\mathbb{R}^d} w^*(x)p(x)x dx + 2c \int_{\mathbb{R}^d} w^*(x)p(x)dx\end{aligned}$$

Set $\nabla_c \mathcal{L} = 0$:

$$c^* \int_{\mathbb{R}^d} w^*(x)p(x)dx = \int_{\mathbb{R}^d} w^*(x)xp(x)dx \implies c^* = \frac{\int_{\mathbb{R}^d} xw^*(x)p(x)dx}{\int_{\mathbb{R}^d} w^*(x)p(x)dx}$$

This fraction is the expectation of x restricted to the inlier region where $w^*(x) = 1$. Thus, we can write it as an expectation:

$$c^* = \mathbb{E}_{x \sim p(x)}[x \mid w^*(x) = 1] = \mathbb{E}_{x \sim p(x)}[x \mid \|x-c\|_2^2 \leq \lambda]$$

- (e) (10 pts) Robust k -means still requires well-structured datasets where the clusters are sufficiently separated. Let's see why! Consider the one-dimensional case where $x \in \mathbb{R}$ and the data generating distribution $p(x)$ is made up of two overlapping Gaussian distributions:

$$p(x) = \frac{1}{2}\mathcal{N}(x|\mu_1, \sigma^2) + \frac{1}{2}\mathcal{N}(x|\mu_2, \sigma^2)$$

Assume $\Delta = \mu_2 - \mu_1 > 0$, $\sigma > 0$. Here, we'll attempt to find two centroids, c_1^* and c_2^* , that are optimal for our data. Write the explicit integral form for the centroid c_1^* based on your answer to the previous part, assuming $w^*(x) = \mathbb{I}_{\{|x-\mu_1| \leq r\}}$ and $r > 0$. You may use $p(x)$ in your solution.

Then, prove that the bias $c_1^* - \mu_1$ is strictly positive for any finite bounding radius $r > 0$, showing that the hard boundary pulls the centroid away from the true mean. *Hint:* A change of variable may help simplify your integral. Additionally, can you break up your integral into two pieces using a u -substitution? How does the Gaussian density change over squared distance?

Answer: Using the derived expectation formula, the explicit integral form for c_1^* with the given weights is:

$$c_1^* = \frac{\int_{\mu_1-r}^{\mu_1+r} xp(x)dx}{\int_{\mu_1-r}^{\mu_1+r} p(x)dx}$$

To find the bias, subtract μ_1 :

$$c_1^* - \mu_1 = \frac{\int_{\mu_1-r}^{\mu_1+r} xp(x)dx - \mu_1 \int_{\mu_1-r}^{\mu_1+r} p(x)dx}{\int_{\mu_1-r}^{\mu_1+r} p(x)dx} = \frac{\int_{\mu_1-r}^{\mu_1+r} (x - \mu_1)p(x)dx}{\int_{\mu_1-r}^{\mu_1+r} p(x)dx}$$

Because $p(x) = \frac{1}{2}\mathcal{N}(x|\mu_1, \sigma^2) + \frac{1}{2}\mathcal{N}(x|\mu_2, \sigma^2)$, we can distribute the integration over the two components. The first component is $\mathcal{N}(x|\mu_1, \sigma^2)$, which is perfectly symmetric around μ_1 . Thus, $\int_{\mu_1-r}^{\mu_1+r} (x - \mu_1)\mathcal{N}(x|\mu_1, \sigma^2)dx = 0$ due to odd symmetry. The bias numerator is entirely driven by the second Gaussian:

$$c_1^* - \mu_1 = \frac{\frac{1}{2} \int_{\mu_1-r}^{\mu_1+r} (x - \mu_1)\mathcal{N}(x|\mu_2, \sigma^2)dx}{\int_{\mu_1-r}^{\mu_1+r} p(x)dx}$$

We use u-substitution $u = x - \mu_1$, meaning $du = dx$ and limits become $[-r, r]$. The density shifts to $\mathcal{N}(u + \mu_1|\mu_2, \sigma^2) \propto \exp(-\frac{(u+\mu_1-\mu_2)^2}{2\sigma^2}) = \exp(-\frac{(u-\Delta)^2}{2\sigma^2})$. The numerator integral becomes:

$$\propto \int_{-r}^r u \exp\left(-\frac{(u-\Delta)^2}{2\sigma^2}\right) du$$

Split this integral into $[-r, 0]$ and $[0, r]$ and flip the negative side:

$$\begin{aligned} &= \int_0^r u \left[\exp\left(-\frac{(u-\Delta)^2}{2\sigma^2}\right) - \exp\left(-\frac{(-u-\Delta)^2}{2\sigma^2}\right) \right] du \\ &= \int_0^r u \left[\exp\left(-\frac{(u-\Delta)^2}{2\sigma^2}\right) - \exp\left(-\frac{(u+\Delta)^2}{2\sigma^2}\right) \right] du \end{aligned}$$

Since $u > 0$ and $\Delta > 0$, it is clear that $(u-\Delta)^2 < (u+\Delta)^2$. This makes the left exponent less negative (larger) than the right exponent, meaning the term in brackets is strictly positive for all $u \in (0, r]$. Therefore, the integral is strictly positive. Since the denominator (a probability mass) is also strictly positive, the entire bias $c_1^* - \mu_1$ is strictly > 0 .

Mixture of Gaussians

In robust k -means, we implicitly added an extra cluster to contain points past a distance threshold away from any centroid, representing a set of outliers. We also discovered a centroid bias away from the true mean. To rectify these issues, we can use a probabilistic generative model as seen in lecture: the Mixture of Gaussians (MoG). Here, we'll modify the MoG seen in lecture to include an extra $(k+1)$ -th component as a uniform distribution for the outliers. For our total data volume V , our mixture distribution with two multivariate Gaussians:

$$p(x) = \pi_1 \mathcal{N}(x|z=1, \mu_1, \Sigma) + \pi_2 \mathcal{N}(x|z=2, \mu_2, \Sigma) + \pi_{\text{out}} \mathcal{U}(x|z=\text{out})$$

where latent variable z is the cluster assignment of x , $\mathcal{U}(x) = \frac{1}{V}$ over the domain of our data, and our original constraint of $\pi_1 + \pi_2 + \pi_{\text{out}} = 1$ still applies.

- (f) (2 pts) Let $\gamma_1(x) = p(z=1|x)$ represent the posterior probability that point x belongs to the first cluster. Write an explicit equation for $\gamma_1(x)$. For this part, you may write Gaussian densities in the form $\mathcal{N}(\dots|\dots)$.

Answer: Using Bayes' rule, the posterior probability is the prior times the likelihood divided by the marginal probability $p(x)$:

$$\gamma_1(x) = \frac{\pi_1 \mathcal{N}(x | \mu_1, \Sigma)}{p(x)} = \frac{\pi_1 \mathcal{N}(x | \mu_1, \Sigma)}{\pi_1 \mathcal{N}(x | \mu_1, \Sigma) + \pi_2 \mathcal{N}(x | \mu_2, \Sigma) + \frac{\pi_{\text{out}}}{V}}$$

- (g) (5 pts) Prove that for an arbitrary unit vector $v \in \mathbb{R}^d$, as the point $x = \alpha v$ moves infinitely far from the centroids ($\alpha \rightarrow \infty$), the posterior probability $\gamma_1(x)$ decays to zero. Explain why the uniform background term stabilizes the denominator and find which cluster $x = \alpha v$ asymptotically belongs to. *Hint:* what property of positive definite matrices can help?

Answer: The multivariate Gaussian density is given by $\mathcal{N}(x | \mu_k, \Sigma) \propto \exp(-\frac{1}{2}(x - \mu_k)^\top \Sigma^{-1}(x - \mu_k))$. Since covariance matrices Σ are positive definite, their inverse Σ^{-1} is also strictly positive definite. For any positive definite matrix, there exists some minimum eigenvalue $\lambda_{\min} > 0$ such that $y^\top \Sigma^{-1} y \geq \lambda_{\min} \|y\|^2$. Let $y = \alpha v - \mu_k$. As $\alpha \rightarrow \infty$, $\|y\|^2 \rightarrow \infty$, meaning the quadratic form approaches ∞ , and consequently $\exp(-\infty) \rightarrow 0$. Thus, both Gaussian densities $\mathcal{N}(x | \mu_1, \Sigma)$ and $\mathcal{N}(x | \mu_2, \Sigma)$ decay to 0.

Substitute this into the $\gamma_1(x)$ formula from (f):

$$\lim_{\alpha \rightarrow \infty} \gamma_1(\alpha v) = \frac{0}{0 + 0 + \frac{\pi_{\text{out}}}{V}} = 0$$

The uniform background term $\frac{\pi_{\text{out}}}{V}$ prevents the denominator from shrinking to zero, saving the equation from a mathematically undefined $0/0$ form and stabilizing numerical calculation. By the same limit evaluation, the posterior for the outlier cluster asymptotically goes to $\frac{\pi_{\text{out}}/V}{\pi_{\text{out}}/V} = 1$, meaning the distant point asymptotically belongs entirely to the outlier cluster.

10 Watch Me Flow Dat

Another recent method of generative modeling is *flow matching* as a competitive alternative to diffusion. This problem is inspired by Flow Matching for Generative Modeling from ICLR 2023. For an optional visual introduction, we'd highly encourage A Visual Dive into Conditional Flow Matching, although all information required for this problem will be provided here.

Our goal is to map a simple prior distribution $p_0(x) = \mathcal{N}(x|0, I)$ to a complex empirical data distribution $p_1(x) \approx q(x)$. A real world example of this would be creating a mapping between pure Gaussian noise (the prior distribution) to a photographer's image dataset (the complex empirical distribution). Note the subscript here denotes time $t \in [0, 1]$ (e.g. p_1 represents the distribution at "time" $t = 1$).

We can represent the mapping between the densities with a time-dependent vector field $v_t : \mathbb{R}^d \rightarrow \mathbb{R}^d$ that assigns a velocity vector to each data point x at every $t \in [0, 1]$. The ordinary differential equation (ODE) $\frac{dx}{dt} = v_t(x)$ defines the trajectory of a point flowing between the two distributions p_0 and q . Then by integrating the ODE from $t = 0$ to $t = 1$, we can transform a point in the original distribution x_0 to a final point x_1 :

$$x_1 = x_0 + \int_0^1 v_t(x(t)) dt$$

The true marginal vector "field" $v_t(x)$ is analytically intractable, so we try to construct it using a tractable conditional probability path $p_t(x|x_1)$ targeting a specific point in the target distribution $x_1 \sim q(x_1)$. Specifically, we can create a line between any two points in the prior and target distributions $x_t = (1-t)x_0 + tx_1$. For this form, the conditional vector field guiding a point toward target x_1 is

$$u_t(x|x_1) = \frac{x_1 - x}{1 - t}$$

Take a look at the figure showing how we draw lines between points in the original and target distribution. Intuitively, you can think of $u_t(x|x_1)$ as the slope of the line (rise over run!). We treat moving along the x -axis as moving through time and moving along the y -axis as moving through space. Given this conditional vector field, our true marginal vector field will be the expected conditional field, weighted by the posterior probability of the target data point given the current spatial state:

$$v_t(x) = \int u_t(x|x_1)p_t(x_1|x) dx_1$$

The Marginal Field

We first show why $v_t(x)$ has this specific integral form. For a fixed time t and fixed spatial position x , consider an arbitrary vector $v \in \mathbb{R}^d$.

- (a) (2 pt) Write the expected squared error $J(v)$ between v and the conditional vector field $u_t(x|x_1)$ over all possible targets x_1 . *Hint:* what distribution should this expectation be over?

Answer: The expectation is computed over the posterior probability of the target given the current spatial state, which is $p_t(x_1 | x)$.

$$J(v) = \mathbb{E}_{x_1 \sim p_t(x_1|x)} [\|v - u_t(x|x_1)\|_2^2]$$

- (b) (4 pts) Take the gradient of $J(v)$ with respect to v and prove that the minimizer v^* is the exact definition of the marginal vector field $v_t(x)$.

Answer: Substitute the integral form for the expectation and differentiate under the integral:

$$\nabla_v J(v) = \nabla_v \int \|v - u_t(x|x_1)\|_2^2 p_t(x_1|x) dx_1 = \int 2(v - u_t(x|x_1)) p_t(x_1|x) dx_1$$

To find the minimizer v^* , set the gradient to zero:

$$\int 2v p_t(x_1|x) dx_1 - \int 2u_t(x|x_1) p_t(x_1|x) dx_1 = 0$$

$$v^* \int p_t(x_1|x) dx_1 = \int u_t(x|x_1) p_t(x_1|x) dx_1$$

Since $p_t(x_1|x)$ is a valid probability distribution over x_1 , it integrates to 1. Thus:

$$v^* = \int u_t(x|x_1) p_t(x_1|x) dx_1$$

This matches the exact definition of $v_t(x)$ provided earlier.

Equivalent Objectives

Ideally, we would fit a model $v(x, t|\theta)$ to match the marginal vector field:

$$\mathcal{L}_{\text{marginal}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,1), x \sim p_t(x)} [\|v(x, t|\theta) - v_t(x)\|_2^2]$$

However, this objective is intractable since we don't know the solution to $v_t(x)$. We instead optimize the Conditional Flow Matching (CFM) objective, which regresses directly against the tractable conditional fields:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,1), x_1 \sim q(x_1), x \sim p_t(x|x_1)} [\|v(x, t|\theta) - u_t(x|x_1)\|_2^2]$$

- (c) (7 pts) Using the definition of $v_t(x)$ from the previous parts, prove mathematically that optimizing $\mathcal{L}_{\text{CFM}}(\theta)$ is equivalent to optimizing $\mathcal{L}_{\text{marginal}}(\theta)$. Concretely, prove

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathcal{L}_{\text{marginal}}(\theta) + C$$

where C is a constant independent of the model parameters θ . *Hint:* Bayes' theorem tells us $p_t(x|x_1)q(x_1) = p_t(x_1|x)p_t(x)$. You may find the notation $p_t(x, x_1)$ to be slightly confusing; this joint distribution should be interpreted explicitly as $p(x_t = x, x_1 = x_1)$, representing the joint probability that the process is at location x at time t and will end up at location x_1 at time 1. We use $p_t(x, x_1)$ purely for notational convenience.

Answer: First, using the Hint, we express the joint expectation in $\mathcal{L}_{\text{CFM}}(\theta)$ by grouping over x :

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, x \sim p_t(x)} [\mathbb{E}_{x_1 \sim p_t(x_1|x)} [\|v(x, t|\theta) - u_t(x|x_1)\|_2^2]]$$

Expand the squared L2 norm inside the inner expectation by subtracting and adding $v_t(x)$:

$$\begin{aligned} & \|v(x, t|\theta) - v_t(x) + v_t(x) - u_t(x|x_1)\|_2^2 \\ &= \|v(x, t|\theta) - v_t(x)\|_2^2 + 2(v(x, t|\theta) - v_t(x))^\top (v_t(x) - u_t(x|x_1)) + \|v_t(x) - u_t(x|x_1)\|_2^2 \end{aligned}$$

Now we take the expectation of this expanded form over $x_1 \sim p_t(x_1|x)$: The first term is constant with respect to x_1 , remaining $\|v(x, t|\theta) - v_t(x)\|_2^2$. The cross term contains the expected value of $u_t(x|x_1)$:

$$\begin{aligned} & \mathbb{E}_{x_1|x} \left[2(v(x, t|\theta) - v_t(x))^\top (v_t(x) - u_t(x|x_1)) \right] \\ &= 2(v(x, t|\theta) - v_t(x))^\top (v_t(x) - \mathbb{E}_{x_1|x}[u_t(x|x_1)]) \end{aligned}$$

From part (b), we know $\mathbb{E}_{x_1|x}[u_t(x|x_1)] = v_t(x)$. Therefore, $v_t(x) - v_t(x) = 0$, and the cross term completely cancels out. Applying the outer expectation over t and x back to the remaining terms yields:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,x} [\|v(x, t|\theta) - v_t(x)\|_2^2] + \mathbb{E}_{t,x,x_1} [\|v_t(x) - u_t(x|x_1)\|_2^2]$$

The first term is exactly the definition of $\mathcal{L}_{\text{marginal}}(\theta)$. The second term $\mathbb{E}_{t,x,x_1} [\|v_t(x) - u_t(x|x_1)\|_2^2]$ depends only on the true fields and is entirely independent of the model parameters θ . We assign this constant to C . Thus,

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathcal{L}_{\text{marginal}}(\theta) + C$$

MLE on Discrete Flows

We know from the previous part that optimizing $\mathcal{L}_{\text{CFM}}(\theta)$ directly matches the conditional vector fields. Now, we'll discretize the flow into a probabilistic Markov chain so we can maximize its likelihood.

Assume our time t can now be discretized into T steps of size $\Delta t = 1/T$. Let $k \in \{1, \dots, T\}$ be the discrete step index. We observe our data as n target trajectories $\{x_0^{(i)}, \dots, x_T^{(i)}\}_{i=1}^n$, where in each trajectory, we have $x_j^{(i)} \in \mathbb{R}^d$ generated by the true linear optimal transport conditional path $x_t = (1-t)x_0 + tx_1$.

We now model the mapping between our prior distribution and target distribution using a Markov chain, with states occurring at each discrete time-step. Using a Markov chain here is useful as we only need to model the transition probabilities efficiently and can ignore the history of a trajectory. Thus, our model will fit the transition probabilities with a multivariate Gaussian:

$$p_\theta(x_k | x_{k-1}) = \mathcal{N}(x_k | x_{k-1} + \Delta t \cdot v(x_{k-1}, (k-1)\Delta t | \theta), \sigma^2 I_{d \times d})$$

- (d) (4 pts) Write the log-likelihood $\ell(\theta)$ of the observed trajectories. Assume trajectories are independent, and that the process follows a first-order Markov chain assumption where the transition to state k only depends on the previous state $(k-1)$. Expand the product into a sum of multivariate Gaussian log-densities. *Note:* recall the general formula for a d -dimensional multivariate Gaussian density:

$$p(x | \mu, \Sigma) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right)$$

Answer: By independence of trajectories and the first-order Markov assumption, the overall likelihood is the product over all n trajectories and all T time steps. Taking the logarithm turns these products into summations:

$$\ell(\theta) = \sum_{i=1}^n \sum_{k=1}^T \log p_\theta(x_k^{(i)} | x_{k-1}^{(i)})$$

Substituting the multivariate Gaussian density with $\mu = x_{k-1}^{(i)} + \Delta t \cdot v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta)$ and covariance $\Sigma = \sigma^2 I$ (making $|\Sigma| = \sigma^{2d}$ and $\Sigma^{-1} = \frac{1}{\sigma^2} I$):

$$\ell(\theta) = \sum_{i=1}^n \sum_{k=1}^T \left(-\frac{d}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left\| x_k^{(i)} - x_{k-1}^{(i)} - \Delta t \cdot v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta) \right\|_2^2 \right)$$

- (e) (4 pts) Derive the simplified MLE objective. Prove that maximizing this log-likelihood is mathematically equivalent to minimizing the squared Euclidean distance between $v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta)$ and the discrete empirical velocity $\frac{x_k^{(i)} - x_{k-1}^{(i)}}{\Delta t}$:

$$\arg \max_{\theta} \ell(\theta) = \arg \min_{\theta} \sum_{i=1}^n \sum_{k=1}^T \left\| v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta) - \frac{x_k^{(i)} - x_{k-1}^{(i)}}{\Delta t} \right\|_2^2$$

Answer: Starting from the $\ell(\theta)$ expression in (d), we ignore the $-\frac{d}{2}\log(2\pi\sigma^2)$ term since it does not depend on θ . Maximizing the remaining negative term is equivalent to minimizing the positive squared norm:

$$\arg \max_{\theta} \ell(\theta) = \arg \min_{\theta} \sum_{i=1}^n \sum_{k=1}^T \frac{1}{2\sigma^2} \left\| x_k^{(i)} - x_{k-1}^{(i)} - \Delta t \cdot v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta) \right\|_2^2$$

We factor out Δt from inside the norm. Because the scalar comes out of the L2 norm squared as Δt^2 :

$$= \arg \min_{\theta} \sum_{i=1}^n \sum_{k=1}^T \frac{\Delta t^2}{2\sigma^2} \left\| \frac{x_k^{(i)} - x_{k-1}^{(i)}}{\Delta t} - v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta) \right\|_2^2$$

Since $\frac{\Delta t^2}{2\sigma^2} > 0$ is a constant scaling factor, it does not change the optimal parameters θ (the arg min). We drop it, swap the terms inside the norm (since $\|A - B\|_2^2 = \|B - A\|_2^2$), and are left with:

$$= \arg \min_{\theta} \sum_{i=1}^n \sum_{k=1}^T \left\| v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta) - \frac{x_k^{(i)} - x_{k-1}^{(i)}}{\Delta t} \right\|_2^2$$

- (f) (4 pts) Using the linear optimal transport path definition $x_t = (1-t)x_0 + tx_1$, evaluate the discrete empirical velocity. Prove that the MLE objective derived in the previous part strictly recovers the Conditional Flow Matching objective $\mathcal{L}_{\text{CFM}}(\theta)$ when evaluated at the discrete time-steps. *Hint:* Discretize the continuous optimal transport path definition using k . *Hint:* Remember that the true conditional vector field is defined as

$$u_t(x_t | x_1) = \frac{x_1 - x_t}{1 - t}$$

Answer: First, find the spatial coordinates at the discrete time steps. At step k , $t = k\Delta t$, so $x_k = (1 - k\Delta t)x_0 + (k\Delta t)x_1$. At the previous step $t = (k-1)\Delta t$, $x_{k-1} = (1 - (k-1)\Delta t)x_0 + (k-1)\Delta t x_1$. Evaluate the discrete empirical velocity by finding the difference:

$$\frac{x_k - x_{k-1}}{\Delta t} = \frac{-\Delta t x_0 + \Delta t x_1}{\Delta t} = x_1 - x_0$$

Now, let us evaluate the true continuous conditional vector field at the point x_{k-1} and time $(k-1)\Delta t$:

$$u_{(k-1)\Delta t}(x_{k-1} | x_1) = \frac{x_1 - x_{k-1}}{1 - (k-1)\Delta t}$$

Substitute x_{k-1} :

$$\begin{aligned} x_1 - x_{k-1} &= x_1 - [(1 - (k-1)\Delta t)x_0 + (k-1)\Delta t x_1] \\ &= (1 - (k-1)\Delta t)x_1 - (1 - (k-1)\Delta t)x_0 = (1 - (k-1)\Delta t)(x_1 - x_0) \end{aligned}$$

Plugging this back into $u_{(k-1)\Delta t}(x_{k-1} | x_1)$ perfectly cancels out the $(1 - (k-1)\Delta t)$ denominator:

$$u_{(k-1)\Delta t}(x_{k-1} | x_1) = x_1 - x_0$$

Since both the discrete empirical velocity and the true conditional vector field equal $x_1 - x_0$, we can substitute $u_{(k-1)\Delta t}(x_{k-1} | x_1)$ directly into our objective from part (e):

$$\arg \min_{\theta} \sum_{i=1}^n \sum_{k=1}^T \left\| v(x_{k-1}^{(i)}, (k-1)\Delta t | \theta) - u_{(k-1)\Delta t}(x_{k-1}^{(i)} | x_1^{(i)}) \right\|_2^2$$

This discrete double summation over n paths and T steps is exactly the Monte-Carlo sample approximation of the expected value found in $\mathcal{L}_{\text{CFM}}(\theta)$. Thus, evaluating the MLE over these discretized time steps strictly recovers the CFM regression loss.